

## DSA0601- DATA HANDLING AND VISUALIZATION EXPERIMENTS 6 – 10

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### **EXPERIMENT 6**

#### **Product Sales Analysis**

Dataset: Monthly Product Sales

| Product ID | Product Name | January Sales | February Sales | March Sales |
|------------|--------------|---------------|----------------|-------------|
| 1          | Product A    | 2000          | 2200           | 2400        |
| 2          | Product B    | 1500          | 1800           | 1600        |
| 3          | Product C    | 1200          | 1400           | 1100        |

#### **CODE:**

```
# Dataset
```

```
product <- data.frame(  
  Product = c("Product A","Product B","Product C"),  
  January = c(2000,1500,1200),  
  February = c(2200,1800,1400),  
  March = c(2400,1600,1100)  
)
```

```
# 1. Grouped Bar Chart
```

```
sales <- as.matrix(product[,2:4])  
barplot(sales,  
  beside=TRUE,  
  col=c("red","blue","green"),
```

```

names.arg=product$Product,
xlab="Products",
ylab="Sales",
main="Quarter 1 Product Sales") legend("topright",
legend=c("January","February","March"), fill=c("red","blue","green"))

```

## # 2. Stacked Area Chart

```

months <- c(1,2,3)
matplot(months, t(sales), type="l",
        lwd=2,
        col=c("red","blue","green"),
        xaxt="n",
        xlab="Months",
        ylab="Sales",
        main="Sales Trend")
axis(1, at=1:3,
     labels=c("January","February","March"))
legend("topleft",
      legend=product$Product,
      col=c("red","blue","green"),
      lty=1)

```

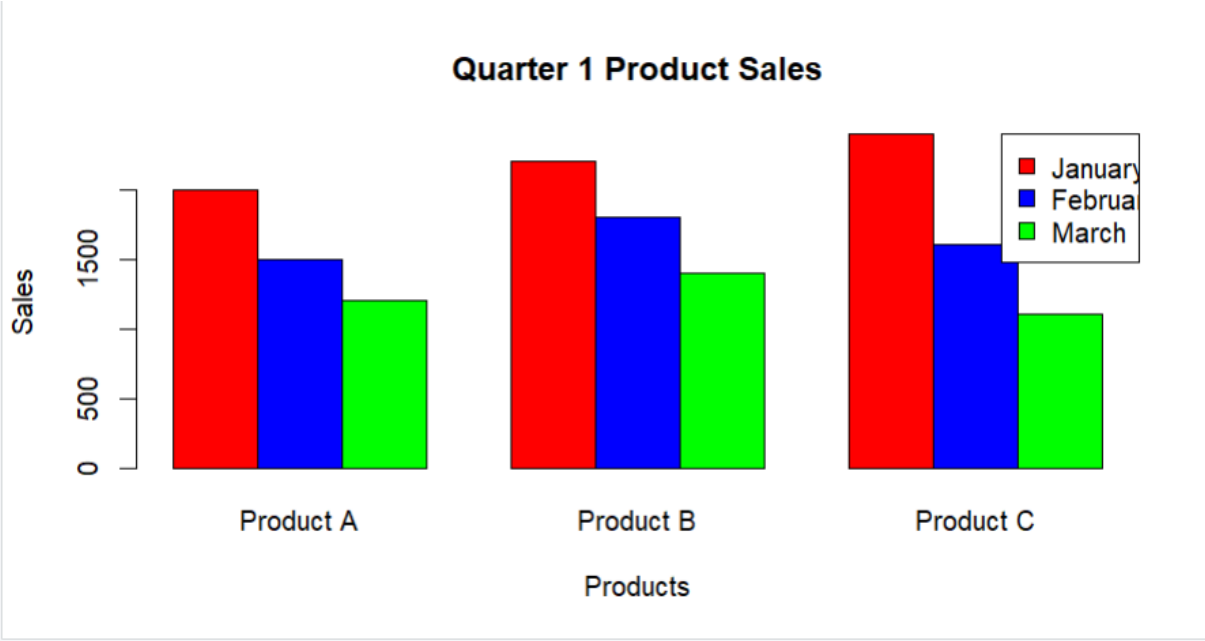
## # 3. Table

Product

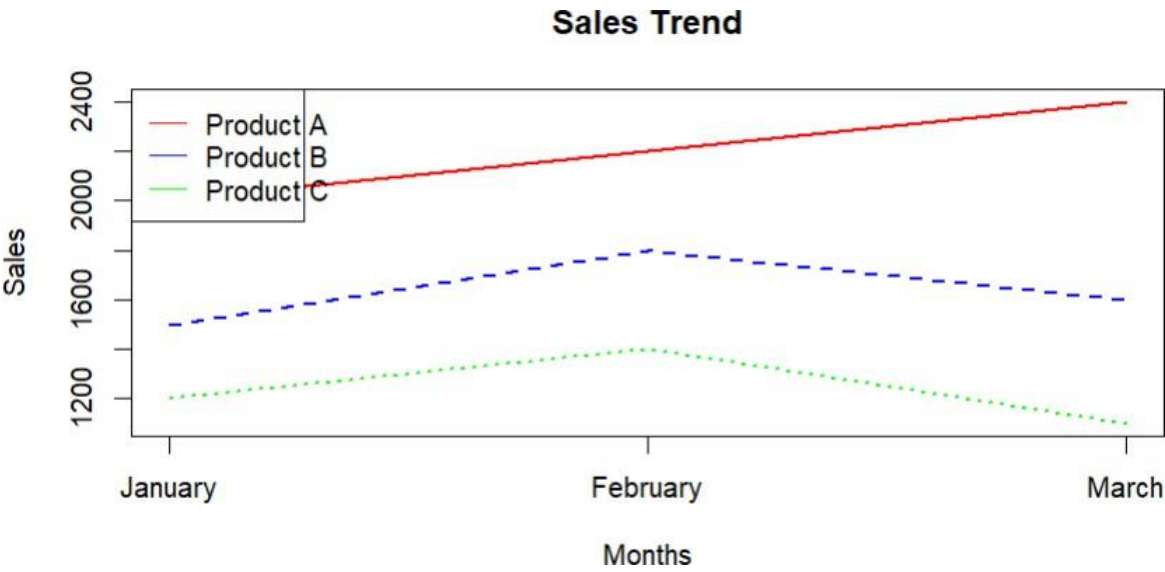
## QUESTION

S:

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.



3. Build a table to show the monthly sales data for each product. Label the table elements.

```
#-----  
# 3. Table  
#-----  
product  
  Product January February March  
Product A   2000    2200   2400  
Product B   1500    1800   1600  
Product C   1200    1400   1100
```

## EXPERIMENT 7

### Customer Demographics Analysis

Dataset: Customer Demographics

| Customer ID | Age | Gender | Income (in \$) |
|-------------|-----|--------|----------------|
| 1           | 28  | Female | 50000          |
| 2           | 35  | Male   | 60000          |
| 3           | 42  | Female | 75000          |

### CODE:

```
# Customer Demographics
customer <- data.frame(
  CustomerID=c(1,2,3),
  Age=c(28,35,42),
  Gender=c("Female","Male","Female"),
  Income=c(50000,60000,75000)
)

# 1. Age Distribution Bar Chart barplot(customer$Age,
  names.arg=customer$CustomerID, col="skyblue",
  xlab="Customer ID",
  ylab="Age",
  main="Customer Age Distribution") #

2. Gender Pie Chart
gender_count <- table(customer$Gender)
```

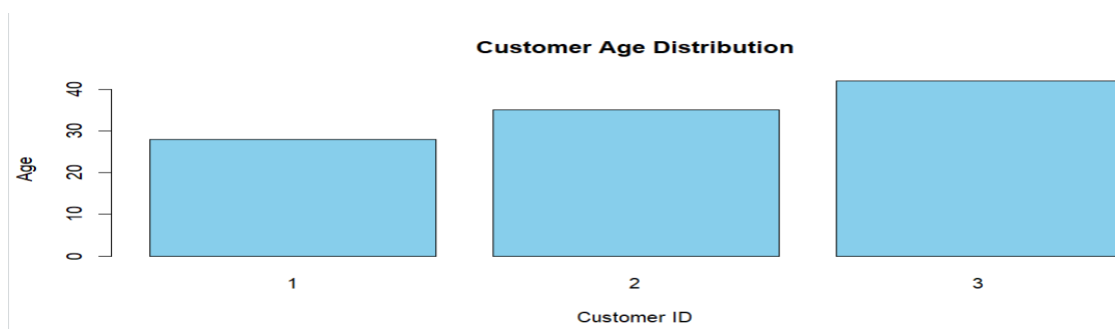
```
pie(gender_count,
    col=c("pink","lightblue"),
    main="Gender Distribution") #
```

### 3. Table

Customer

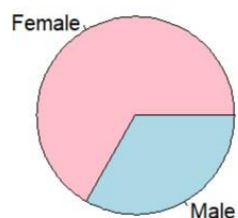
## QUESTIONS

1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.



2. Generate a pie chart to represent the distribution of customers by gender.

**Gender Distribution**



3. Build a table to show the demographic information for each customer. Label the table elements.

```
#-----
# 3. Table
#-----
customer
CustomerID Age Gender Income
1 28 Female 50000
2 35 Male 60000
3 42 Female 75000
```

## EXPERIMENT 8

### Employee Performance Analysis

Dataset: Employee Performance

| Employee ID | Department | Years of Service | Performance Score |
|-------------|------------|------------------|-------------------|
| 1           | Sales      | 5                | 85                |
| 2           | HR         | 3                | 92                |
| 3           | Marketing  | 7                | 78                |

#### CODE:

```
# Employee Performance
employee <- data.frame(
  EmployeeID=c(1,2,3),
  Department=c("Sales","HR","Marketing"),
  Years=c(5,3,7),
  Performance=c(85,92,78)
)

# 1. Line Chart
plot(employee$Years,
      employee$Performance,
      type="o",
      pch=16,
      col="blue",
      xlab="Years of Service", ylab="Performance
      Score", main="Employee Performance Trend")

# 2. Department Bar Chart
```

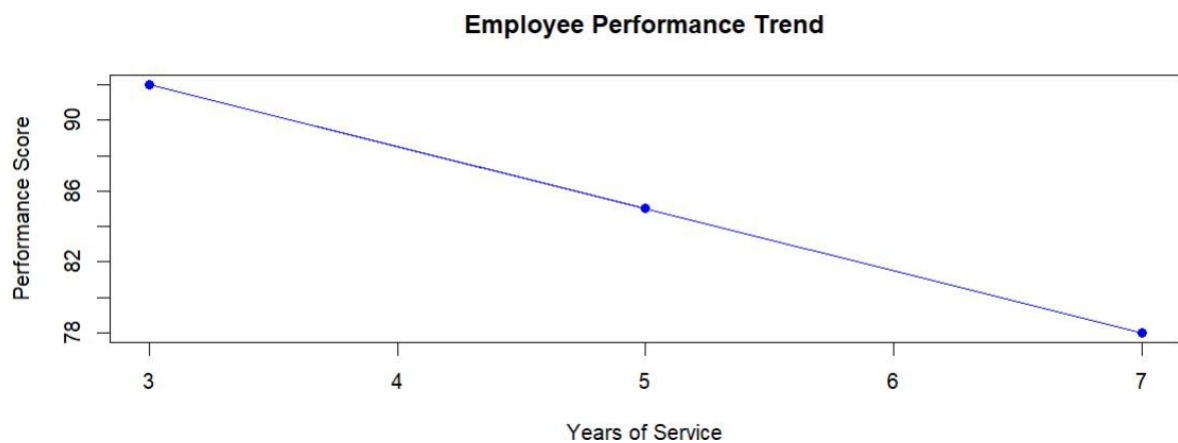
```
dept <- table(employee$Department) barplot(dept,
col="orange", xlab="Department",
ylab="Number of Employees",
main="Employees by Department") #
```

3. Table

Employee

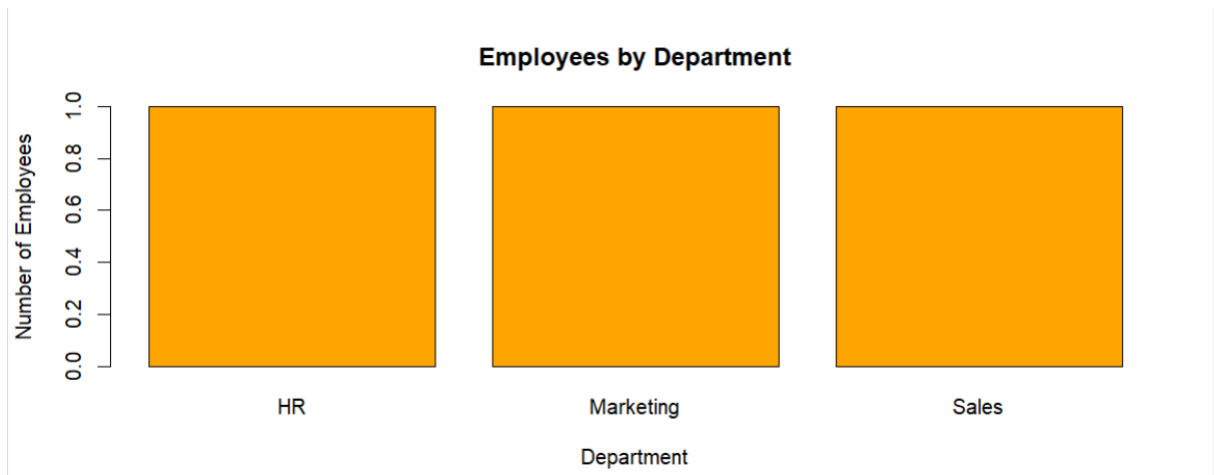
### QUESTIONS:

1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.



2. Generate a bar chart showing the distribution of employees across different departments.

Label the chart elements.



3. Build a table to display the performance data for each employee. Label the table elements.

```
#-----
# 3. Table
#-----
employee
EmployeeID Department Years Performance
      1      Sales      5          85
      2        HR      3          92
      3 Marketing      7          78
```

## EXPERIMENT G

### Product Inventory Management

Dataset: Product Inventory



| Product ID | Product Name | Quantity Available | Price (in \$) |
|------------|--------------|--------------------|---------------|
| 1          | Product A    | 250                | 20            |
| 2          | Product B    | 175                | 15            |
| 3          | Product C    | 300                | 18            |

## CODE

# Product Inventory

```
inventory <- data.frame(
  Product=c("Product A","Product B","Product C"),
  Quantity=c(250,175,300),
  Price=c(20,15,18)
)
```

```
# 1. Quantity Bar Chart barplot(inventory$Quantity,
  names.arg=inventory$Product,
  col="green",
  xlab="Products",
  ylab="Quantity",
  main="Inventory Quantity")
```

# 2. Stacked Bar Chart

```
data <- rbind(inventory$Quantity,
  inventory$Price)
barplot(data,
```

```

col=c("blue","red"),
names.arg=inventory$Product, xlab="Products",
ylab="Values",
main="Quantity and Price")
legend("topright",
      legend=c("Quantity","Price"),
      fill=c("blue","red"))

```

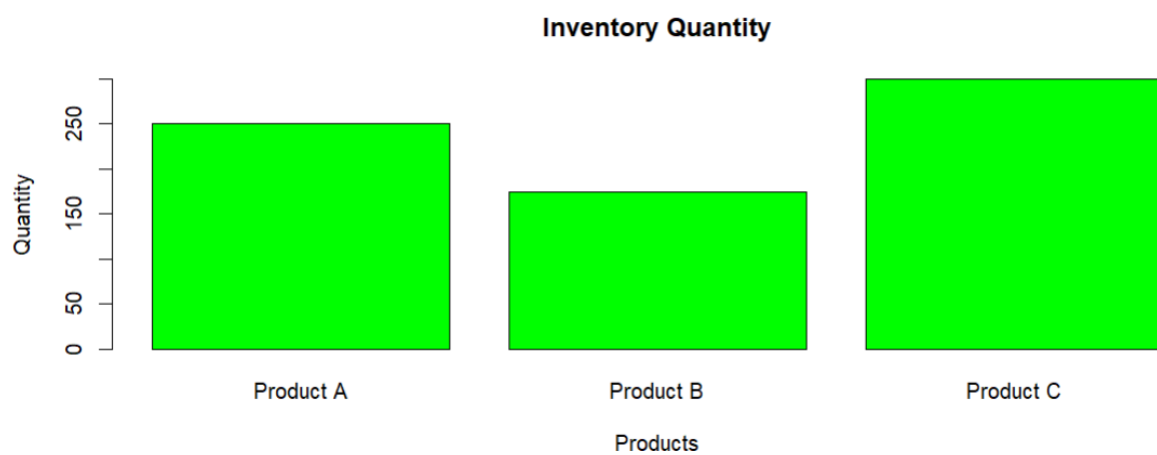
# 3. Table

Inventory

## QUESTION

S:

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.



2. Generate a stacked bar chart to show the quantity of each product within different product categories.



3. Build a table to show the inventory data for each product. Label the table elements.

```
#-----
# 3. Table
#-----
inventory
  Product Quantity Price
Product A      250    20
Product B      175    15
Product C      300    18
```

## EXPERIMENT 10

### Survey Responses Analysis

Dataset: Survey Responses

| Survey ID | Question 1 | Question 2 | Question 3 |
|-----------|------------|------------|------------|
| 1         | A          | B          | C          |
| 2         | B          | A          | D          |
| 3         | C          | A          | B          |

### CODE:

```
# Survey Responses survey
<- data.frame(
  SurveyID=c(1,2,3),
  Q1=c("A","B","C"),
```

```

Q2=c("B","A","A"),
Q3=c("C","D","B")
)
# 1. Grouped Bar Chart (Q1) q1
<- table(survey$Q1) barplot(q1,
  col=c("red","blue","green"),
  xlab="Responses",
  ylab="Frequency",
  main="Question 1 Responses") #

```

## 2. Stacked Bar Chart

```

responses <- rbind(
  table(factor(survey$Q1, levels=c("A","B","C","D"))),
  table(factor(survey$Q2, levels=c("A","B","C","D"))),
  table(factor(survey$Q3, levels=c("A","B","C","D")))
)
barplot(t(responses),
  beside=FALSE,
  col=c("red","blue","green"),
  xlab="Responses",
  ylab="Count",
  main="Survey Responses")
legend("topright",
  legend=c("Question1","Question2","Question3"),
  fill=c("red","blue","green"))

```

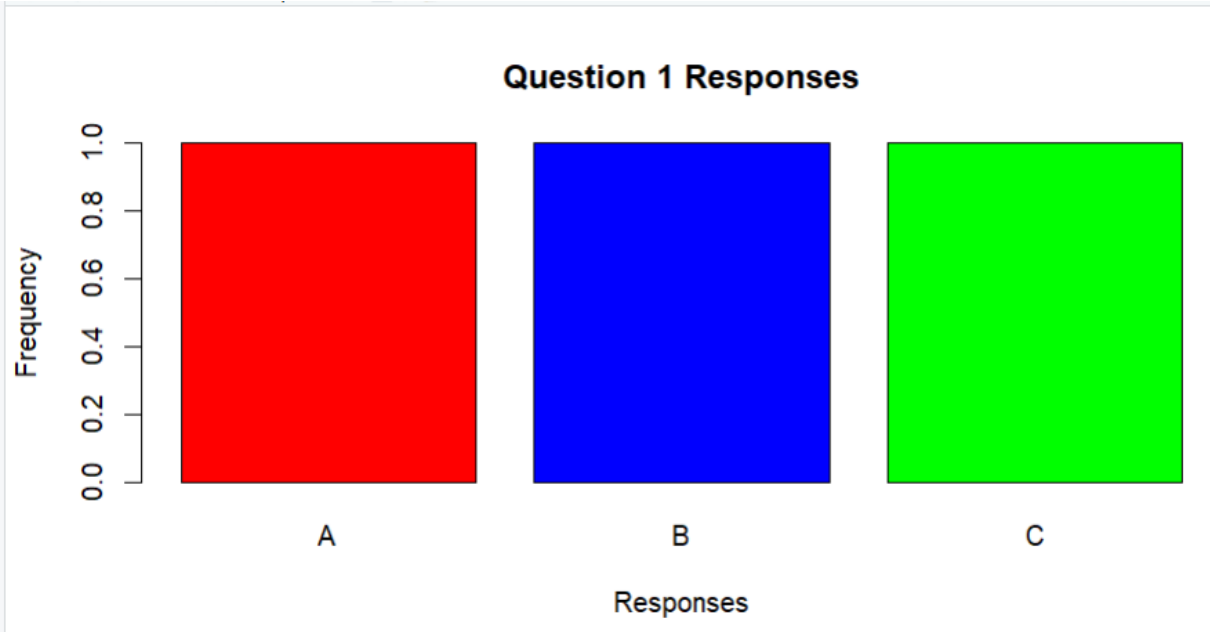
## # 3. Table

survey

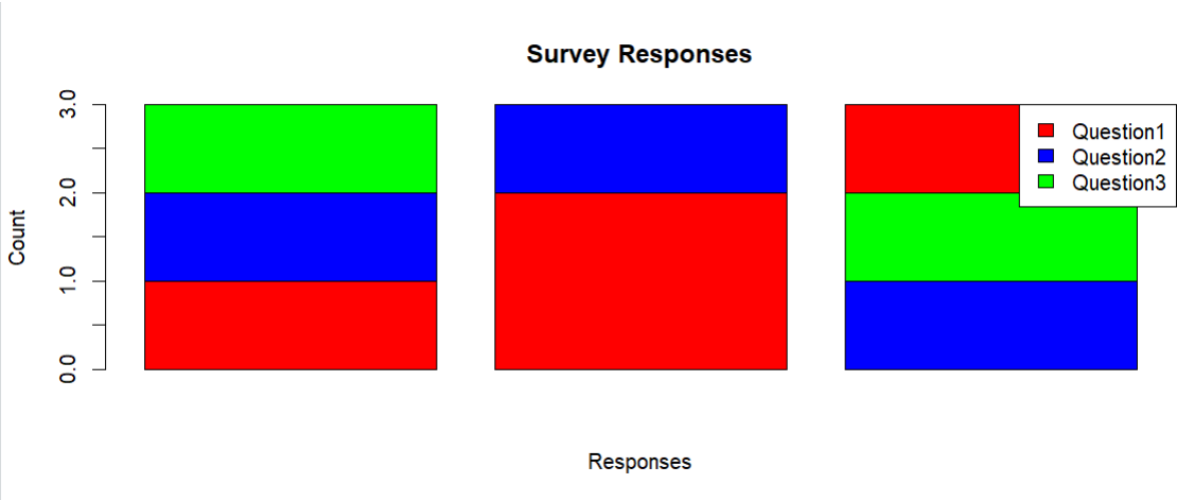
## QUESTION

S:

1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.



2. Generate a stacked bar chart to represent the overall distribution of responses for all three questions.



3. Build a table to show the survey response data for each survey. Label the table elements.

```
#-----  
# 3. Table  
#-----  
survey  
SurveyID Q1 Q2 Q3  
1 A B C  
2 B A D  
3 C A B
```